

Getting Started With AI Agents in HR

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Initiatives: Transform HR through AI; Technologies and Markets

Given their potential business impact for HR, AI agents have become one of the most discussed trends in HR technology. This research helps CHROs identify high-impact use cases, technologies and pilot strategies to confidently start their AI agent journey and achieve measurable results.

Insights at a Glance

AI agents are poised to fundamentally transform HR workflows and services to the organization, with projections suggesting they will automate or perform 50% of current HR-related activities by 2030. However, despite market hype and abundant vendor offerings, the real-world ability of HR functions to implement and extract value from them lags significantly. While 46% of HR functions reported in 2025 a plan to implement agents in the coming year, ¹ HR functions risk rushing into technical pilots that lack clear business cases or attempting to deploy agents where they're not yet viable.

Gartner categorizes AI agent use cases across a spectrum of complexity and risk:

- **AI agent “sweet spot”** — Medium-complexity tasks where the environment is too dynamic for rigid automation, yet the cost of error is low enough to manage risk
 - **Recommendation for CHROs:** This is the primary entry point for pilots. Focus here to demonstrate adaptability and achieve measurable outcomes.
- **AI agents are overkill** — Routine tasks that follow a fixed sequence of steps and don't require dynamic planning
 - **Recommendation for CHROs:** Rely on mature, traditional automation technologies, which are often more efficient for these tasks.

- **AI agents are not ready, or the risks are too high** — Tasks where current agent reliability is insufficient or the potential consequences are too great
 - **Recommendation for CHROs:** Hold off on deployment. These areas require human judgment and should not be automated until the technology matures.

Strategic Planning Assumption

By 2030, 50% of current HR activities will be AI-automated or performed by AI agents, fundamentally transforming HR's work, roles and workflows.

Impact

Interest in AI agents is surging across HR, with 46% of HR functions reporting in 2025 they plan to implement semiautonomous or fully autonomous agents in the coming year. ¹ In addition, Gartner reports a 2,000% year-over-year increase in HR AI-agent-related inquiries for 2025. This dramatic spike not only underscores the intense market hype and rising expectations but also signals a growing need for deeper strategic guidance and practical support. This surge reflects both excitement about AI's potential and uncertainty about how to translate that potential into meaningful HR outcomes.

Many HR functions rush into pilots or implementations without a clear understanding of where AI agents truly add value. Organizations risk wasting investments, derailing regulatory compliance, creating poor employee experiences or missing opportunities for impact. The most successful pilots focus on delivering measurable business outcomes and reengineering HR work, rather than simply showcasing technical capabilities. CHROs must work with their team and enterprise AI center of excellence (or CIO and IT) to carefully prioritize where to start to ensure sustainable impact and avoid costly missteps.

Actions

To get started with AI agents, partner with your HR technology leader and/or IT to:

- Identify and prioritize the right high-impact use case(s), focusing on an area with significant business value, especially where the environment is too dynamic for traditional automation and the cost of errors is low.
- Choose technology based on fit, not complexity, letting the requirements of each use case guide technology selection, rather than opting for the most sophisticated or advanced solution.

- Design pilots for measurable business outcomes, incorporating clear success metrics and feedback mechanisms.

Cautions

- **Don't underestimate data quality:** Poor or incomplete data can produce unreliable or biased results, jeopardizing both outcomes and trust in AI-driven HR decisions.
- **Account for the total cost of ownership:** Beyond licensing fees, factor in hidden costs such as pricing models, integration, ongoing maintenance, staff training, and compliance or data privacy requirements.
- **Design for user adoption:** Successful AI agent deployment depends on intuitive interfaces, clear communication, a human-first approach and robust training for all users — HR teams, employees and candidates alike. Ensure you implement effective change management to support adoption and ongoing engagement.

How to Execute

To maximize the impact of AI agents in HR, guide your team in taking a strategic, step-by-step approach. This begins with identifying and prioritizing the most valuable use cases, selecting technology solutions that align with business objectives and running pilots that focus on delivering measurable business outcomes. The following sections outline key actions and considerations for each stage of this journey.

AI agents are autonomous or semiautonomous software entities that use AI techniques to perceive, make decisions, take actions and achieve goals in digital or physical environments. The terms "AI agents" and "agentic AI" are often used interchangeably in the market. However, Gartner sees them as distinct, with agentic AI being a broader approach to building AI solutions based on the use of software entities that are classified completely or at least partially as AI agents.

Prioritize the Right High-Impact AI Agent Use Case

Selecting the right initial use case(s) is an important first step to realizing the full value of AI agents in HR. Targeted deployments ensure resources are focused where they can deliver the greatest impact, accelerating adoption and measurable business outcomes. For further context on HR AI agents, see [Quick Answer: AI Agent Essentials for CHROs](#).

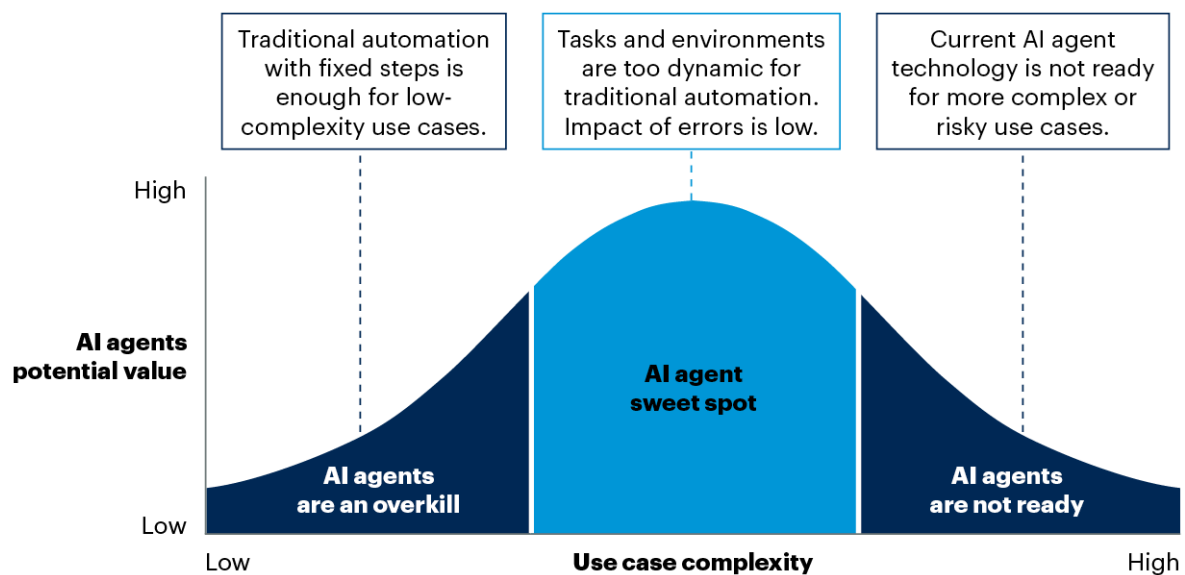
Unlike traditional automation, AI agents bring flexibility and autonomy, making them ideal for HR processes that are complex and variable and require adaptive problem solving. However, not every HR process is an immediate fit. The most successful use cases are those that:

- Address a well-defined, high-impact business problem
- Resolve bottlenecks in human workflows
- Allow agents to demonstrate adaptability and autonomy
- Provide clear, measurable results that matter to HR and the business

To maximize value and adoption, prioritize the sweet spot — where AI agent capabilities best support your organization’s key goals (see Figure 1).

Figure 1: The AI Agent Sweet Spot

The AI Agent Sweet Spot Illustrative



Source: Gartner
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Gartner

Table 1 outlines a range of use cases by complexity to help you identify the most suitable starting points. Use the examples as a guide, but prioritize use cases that address your most critical HR outcomes.

Table 1: AI Agent Use-Case Categories and Examples

(Enlarged table in Appendix)

Use-case type	Description	Examples
AI agents are overkill	Low-complexity use cases where tasks can be completed through a fixed sequence of steps, without the need for dynamic planning More mature technologies, such as traditional automation or journeys, are typically sufficient and more efficient than deploying AI agents.	■ Onboarding checklist tracking ■ Routine absence requests ■ Standard employee personal information changes
AI agent sweet spot	Medium-complexity tasks — where the work or environment is too dynamic for rigid, step-by-step automation and instead benefits from AI agents' adaptive planning capabilities The risks and consequences are manageable, enabling HR to gain experience, achieve improvements and build confidence before addressing higher-stakes scenarios.	■ Answering and actioning complex, multistep HR policy questions ■ Supporting managers with tailored coaching or feedback ■ Analyzing interview transcripts against job requirements and across candidates for the best fit
AI agents aren't ready, or the risks are too high	High-complexity, high-risk use cases where either AI agent technology isn't ready or the potential risks are too great Current limitations in reliability and dependability make it advisable to hold off on deploying AI agents in these areas until the technology matures or the risk profile changes.	■ Handling sensitive employee relations or disciplinary cases ■ Making final decisions on hiring, promotions, or terminations ■ Advising on compliance or legal matters

Source: Gartner (February 2026)

Align Technology Selection With Business Goals

HR teams have a spectrum of buy-to-build options for AI agents — from prebuilt solutions in HCM suites and HR tech point solutions to no- or low-code platforms and fully custom agent development (see Figure 2). While prebuilt agents offer speed and simplicity, moving beyond them to more customizable solutions typically requires dedicated resources with advanced technical skills. HR and IT should partner together to ensure each use case is matched with the right approach. To maximize impact, let your specific use case and business goals guide the technology selection process, ensuring your choices align with HR priorities and your organization's broader technology strategy.

Figure 2: Emerging AI Agent Landscape

Emerging AI Agent Landscape



Source: Gartner
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Gartner

- **Prebuilt agents** are off-the-shelf, tech-vendor-delivered AI agents. Many HR tech vendors offer these AI agents, which HR functions can readily use within the vendor's application. Vendors design these agents for targeted HR use cases, such as recruitment and employee self-service, without requiring technical expertise. Prebuilt agents are best suited for less complex use cases or those where customization isn't required. Examples: Out-of-the-box agents from HCM platforms, HR service management and other HR technology point solutions
- **No-code agent builders** are technology products that let HR or IT teams create, launch, and manage AI-powered agents without writing code. Vendors may build these tools into their HCM or HR point solutions or offer them as part of other enterprise applications. **No-code platforms** make it possible to design AI agent workflows that address your organization's unique HR needs, offering more customization than prebuilt agents. However, while they don't require coding, these tools still call for some technical know-how and an understanding of development best practices. HR teams with technically skilled members or dedicated technologists achieve the best results with these tools, which require strong governance to ensure data security and compliance. Examples: Custom agent studios within HCM and HR point solutions, n8n and Zapier

- The final two categories, **AI agent development platforms** and **agent training platforms**, require extensive IT capabilities such as coding and AI expertise. While some CHROs have worked with their IT partners to build internal solutions, this scenario is currently less common and more likely in organizations with high AI maturity.

As you narrow down your options, focus on the practical requirements of each solution. Map the complexity of your HR use cases to the capabilities of available technologies, and identify any gaps in technical expertise or IT support that could impact implementation. Prioritize solutions that not only fit your immediate needs but also integrate smoothly with your organization's existing systems, security, risk tolerance and compliance standards, ensuring long-term success and scalability.

Implement and Evaluate Pilot Programs

For CHROs, a successful AI agent pilot is not about simply confirming that the technology functions — it's about demonstrating real business value. Too often, organizations run technical pilots without considering the broader impacts or enable AI agents just because they're available. This narrow approach risks delivering only incremental improvements and overlooks the opportunity for meaningful transformation.

To avoid these pitfalls, it's essential to follow a structured pilot process that keeps business relevance and long-term impact at the forefront:

- **Phase 1** — Start by assembling a cross-functional **fusion team** of HR, IT, AI and subject matter experts who can rapidly iterate agent design and integrate it into workflows. Prioritize use cases that are low-risk and high-value, mapping out processes where flexible automation can make a tangible difference.
- **Phase 2** — As you move forward, clearly define success metrics that align with your desired business outcomes, such as improved efficiency or employee satisfaction, and establish a baseline for measurement. Proactively identify and address potential risks, including data privacy, compliance and bias, with a robust mitigation plan. Engage both end users and those responsible for managing the AI agent piloted in the initiative, providing training and support to ensure confident adoption and effective **performance management**.
- **Phase 3** — Finally, monitor the pilot closely, using feedback and metrics to adjust your approach and make informed decisions about scaling or refining use cases.

By following these steps, you can ensure your AI agent pilot is not just a technical exercise but a strategic initiative that drives real value for HR and the broader organization.

Success Measures

Defining what success looks like from the outset enables HR functions to measure impact, inform investment decisions and guide ongoing adoption. Example success measures include:

Business outcome improvement

- Reduction in HR case resolution time (e.g., the average time to resolve employee inquiries decreases by X%)
- Increase in HR process efficiency (e.g., the number of transactions or requests handled per HR FTE increases)
- Reduction in HR operating costs (e.g., the cost per HR transaction or case drops by X%)

Employee experience and satisfaction

- Improvement in employee satisfaction scores related to HR services (e.g., postinteraction surveys, Net Promoter Score)
- Increase in self-service adoption rates (e.g., the percentage of employees using AI-agent-powered self-service tools)
- Decrease in repeat inquiries or escalation rates

User adoption and engagement

- HR staff and manager adoption rates (e.g., the percentage actively using AI agent tools)
- Completion rates for AI-agent-driven tasks (e.g., onboarding, benefits enrollment)

Quality and accuracy

- Accuracy of AI agent responses (e.g., the percentage of correct or relevant answers provided)
- Reduction in error rates or compliance incidents in HR transactions

Evidence

¹ **Benchmark With Gartner: Mitigating Talent Strategy Risk Amid U.S. Policy Shifts & Tackling Skill Gaps.** This live polling webinar was designed to facilitate discussions among HR leaders about the positions their peers are taking on pressing issues that affect crucial talent decisions. The main topics covered in this webinar include the labor market climate outlook update for talent-based leading indicators of economic conditions, the implications of U.S. policy and market shifts on talent strategy, **agentic AI**, skills-based talent management, and skills gaps. The webinar was conducted on 21 May 2025, with responses from over 30 HR leaders participating across a spectrum of industries. Disclaimer: The results of this survey do not represent global findings or the market as a whole, but rather reflect the sentiments of the respondents and the companies surveyed.

Recommended by the Authors

Some documents may not be available as part of your current Gartner subscription.

[Address 5 Gaps to Unlock HR AI Agent Success](#)

[Prepare for the Future of AI Agents in HR](#)

[Framework for Evaluating AI Capabilities in HCM Suites](#)

[3 Steps to Establish AI-Ready HR Data](#)

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AI agents aren't ready, or the risks are too high

High-complexity, high-risk use cases where either AI agent technology isn't ready or the potential risks are too great
Current limitations in reliability and dependability make it advisable to hold off on deploying AI agents in these areas until the technology matures or the risk profile changes.

- Handling sensitive employee relations or disciplinary cases
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